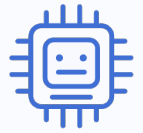


Uranus: Building the Next-Generation Simulation Infrastructure for Embodied AI



Data-driven

No labor-intensive simulator building



Scalable

Support large-scale data generation



Safe & Cost-effective

Reduce real-world interaction cost

Online Inputs (t)



History and current observations



Robot Actions



Multi-view Cameras
(Extrinsics & Intrinsics)



Optional Language

"pick up the red block"

Uranus

Autoregressive Diffusion Model

Prefill

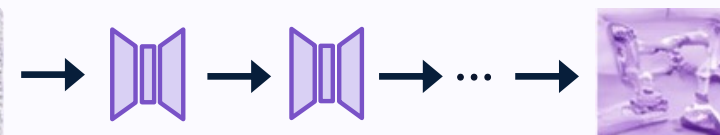
Encode history context and store into KV Cache



Decode: Flow Matching



Gaussian Noise



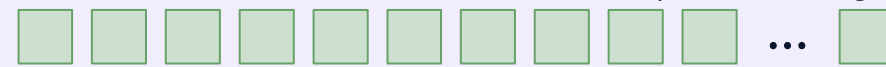
Predicted Latent Frame (t+1)

Append

Append new latent to KV Cache



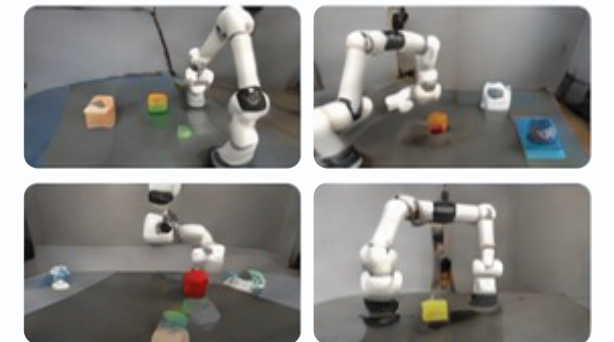
KV Cache (efficient long-horizon memory)



Autoregressive Streaming Loop

Output (t+1)

One latent frame
→ Four RGB frames
(synchronized multi-view)



Cam 1



Cam 2



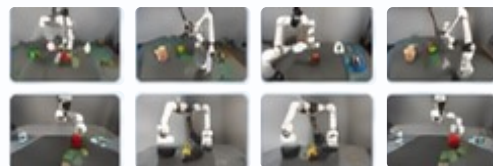
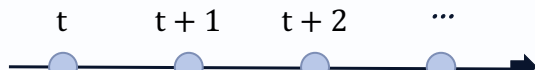
Cam 3



Cam 4

Streaming, Open-ended Rollout

Receive actions online and autoregressively generate one latent frame per step



Low-latency Generation

After inference optimization, Uranus achieves **24 FPS** real-time interactive rollout.



Scalable, Extensible Robot Control

Unified interface for synchronized multi-view generation across diverse robot embodiments and camera configurations.



Multi-view & Flexible Configuration

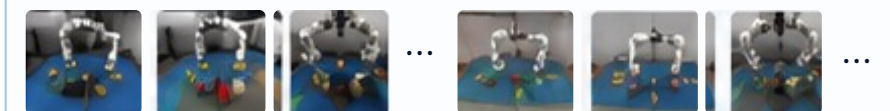


Comprehensive Evaluation

Extensive quantitative and qualitative evaluation on in-distribution and out-of-distribution data.

In-Distribution

Out-of-Distribution



Objective assessment & clear identification of limitations.

Open Source Release: Code, weights, and tools to empower the community.